Vector Spaces IV: Quotient Spaces and Counting over Finite Fields

Problem sheet for the Day 5 class · CSIR NET previous year questions and exam-pattern problems

Name: Date: Score: / 10

Instructions. 10 problems from CSIR NET, GATE (MA) and IIT JAM. A tag with an exam and year marks an actual previous-year question; a tag reading *Exam pattern* marks a problem written in the same style. MCQ: one correct option. MSQ: one or more correct options. NAT: write the number. Suggested time: 30 minutes. **The answer key is posted 24 hours after this sheet.**

1. CSIR NET Dec 2016 · MCQ

What is the number of non-singular 3×3 matrices over $\mathbb{F}_2$, the finite field with two elements?

(A) 168
(B) 384
(C) 2^3
(D) 3^2

2. CSIR NET Dec 2015 · MCQ

Let S denote the set of all the prime numbers p with the property that the matrix $\begin{pmatrix} 91 & 31 & 0 \\ 29 & 31 & 0 \\ 79 & 23 & 59 \end{pmatrix}$ has an inverse in the field $\mathbb{Z}/p\mathbb{Z}$. Then

(A) $S = \{31\}$
(B) $S = \{31, 59\}$
(C) $S = \{7, 13, 59\}$
(D) S is infinite

3. CSIR NET Dec 2014 · MSQ

The matrix $A = \begin{pmatrix} 5 & 9 & 8 \\ 1 & 8 & 2 \\ 9 & 1 & 0 \end{pmatrix}$ satisfies:

(A) A is invertible and the inverse has all integer entries
(B) $\det(A)$ is odd
(C) $\det(A)$ is divisible by 13
(D) $\det(A)$ has at least two prime divisors

4. Exam pattern · NAT

Let $V = M_5(\mathbb{R})$ and W the subspace of symmetric matrices. Then $\dim(V/W) =$ _____.

5. Exam pattern · NAT

Let W be a 2-dimensional subspace of $\mathbb{F}_3^4$. The number of elements of $\mathbb{F}_3^4/W$ is _____.

6. Exam pattern · NAT

The number of 2×2 matrices over $\mathbb{F}_3$ with determinant 1 is _____.

7. Exam pattern · NAT

The number of 1-dimensional subspaces of $\mathbb{F}_3^3$ is _____.

8. Exam pattern · NAT

The number of 2-dimensional subspaces of $\mathbb{F}_2^4$ is _____.

9. Exam pattern NAT The number of 2×2 matrices A over $\mathbb{F}_2$ with $A^2 = 0$ is _____.

10. Exam pattern MSQ Let V be the real polynomials of degree at most 3 and $W = \{p \in V : p(1) = 0\}$. Which of the following are true?

(A) $\dim(V/W) = 1$

(B) $p + W = q + W$ iff $p(1) = q(1)$

(C) $V/W \cong \mathbb{R}$

(D) $\dim W = 2$